1 Building Your First Class and Object



KEY IDEAS

- A **class** is a blueprint; an **object** (instance) is one concrete thing built from it. Calling the class, `Dog("Rex", 3)`, creates an instance.
- Naming: `PascalCase` for classes, `snake_case` for methods and attributes.
- **Instance attributes** (`self.x`) belong to one object; **class attributes** are shared by all instances.

EXAMPLE

```
class Dog:
    species = "Canis familiaris"     # class attribute

    def __init__(self, name, age):   # initializer
        self.name = name             # instance attributes
        self.age = age

    def speak(self):                 # instance method
        return f"{self.name} says Woof!"

rex = Dog("Rex", 3)                  # create an object
print(rex.speak())                   # → Rex says Woof!
print(rex.species, rex.age)          # → Canis familiaris 3
print(isinstance(rex, Dog))          # → True
print(type(rex).__name__)            # → Dog
```